

Evaluating the Quality of the LAION-2B Dataset: Insights into Large-Scale Image Dataset Assessment

Jerome Agius

Supervisor: Dr Dylan Seychell

University of Malta

Msida, Malta

jeromeagius0@gmail.com

Abstract

Generative Artificial Intelligence (AI) systems have enabled large-scale image creation, bringing advanced image generation capabilities within reach of non-specialist users. However, the development of these systems depends heavily on massive web-scraped training datasets such as LAION-5B, with this research focusing on its re-released version, Re-LAION-5B, and the related Stable Diffusion v1.5 model. Due to the nature of web-scraped training data, concerns have increased regarding the ethical implications of these datasets, particularly in relation to bias and unintended correlations embedded within the training corpus. Specifically, non-semantic or spurious features present within the training dataset may serve as shortcuts in downstream tasks, negatively impacting model behaviour and fairness outcomes. While these shortcuts can emerge across many tasks, demographic attributes represent a particularly concerning case, especially when predictions rely on features unrelated to the inherent semantic content of the image.

This thesis investigates demographic bias within the Re-LAION-5B dataset and in generative images produced by Stable Diffusion v1.5, with a direct focus on identifying spurious features in both sources and determining whether the generative model amplifies, alters, or mitigates the patterns observed in the dataset. The study adopts a classifier-based approach derived from prior research, training discriminative models to distinguish between Re-LAION-5B images and Stable Diffusion v1.5 generated images. A set of transformations is introduced to systematically reduce the semantic information available to the classifier and force reliance on specific low-level image features, thereby exposing dataset-specific spurious patterns that can be exploited for image source identification.

Building upon this classifier-based analysis, the images used in the discrimination task are further annotated according to gender and skin-tone attributes. This secondary analysis investigates whether the previously identified dataset-specific spurious features correlate with these sensitive demographic attributes, thereby identifying potential unintended shortcuts for demographic classification. To further contextualise these findings and assess the real-world implications of the observed biases, the demographic distributions of retrieved and generated images across 200 occupation categories are compared against real-world demographic statistics, revealing patterns of representation bias and potential amplification by the generative model.

Finally, based on the spurious features identified in the preceding analyses, this thesis implements and evaluates a combined set of inference-time interventions to debias Stable Diffusion v1.5 without requiring full model retraining. These interventions reduce spurious demographic correlations across low-level, geometric, and semantic

representations, showing that inference-time debiasing can produce measurable and substantive departures from the demographic and spurious feature profile of standard generation. By explicitly linking dataset characteristics, spurious feature learning mechanisms, and generative bias outcomes, this work contributes methodological insights for evaluating large-scale multimodal datasets and practical interventions for developing fairer text-to-image generation systems.

Keywords

Re-LAION-5B, Stable Diffusion, spurious features, demographic bias, dataset auditing, text-to-image generation

1 Introduction

Text-to-image diffusion models, Stable Diffusion v1.5 (SDv1.5) [17] in particular, have achieved widespread deployment while relying on massive web-scraped corpora such as Re-LAION-5B [19], whose automated construction introduces systematic biases that reflect societal inequities [2]. Existing research documents demographic bias in generative outputs [1, 10] but treats models as black boxes, ignoring the *spurious features* — low-level non-semantic characteristics such as colour statistics, texture patterns, edge distributions, frequency domain properties, and compression artefacts — that correlate with demographic attributes and enable biased shortcuts [12, 22]. This thesis addresses these gaps through a mechanistic investigation of spurious features in Re-LAION-5B and SDv1.5, applying established methodologies to expose low-level features that distinguish dataset sources and correlate with demographic attributes, then evaluating targeted inference-time interventions without model retraining. The analysis reveals that Re-LAION-5B and SDv1.5 encode demographic information through qualitatively different mechanisms, with direct consequences for bias attribution and mitigation. Five objectives structure the work: (O1) dataset acquisition and annotation across 200 ISCO-08-aligned occupations; (O2) spurious feature detection via classifier-based transformation analysis; (O3) demographic correlation assessment and occupational representation comparison against real-world statistics; (O4) inference-time debiasing of SDv1.5; and (O5) synthesis of how spurious features and demographic biases are transformed by the SDv1.5 generative process.

2 Background & Literature Review

Re-LAION-5B comprises approximately 5.85 billion image-text pairs harvested via Common Crawl [19], re-released with improved safety filtering [9]; LAION-Aesthetics v2 5+, a quality-filtered offshoot of LAION-2B-en, constitutes the training corpus for SDv1.5 [18].

SDv1.5 is a Latent Diffusion Model [17] that denoises in a compressed latent space via a DDPM U-Net conditioned on CLIP text embeddings [6, 16]; higher classifier-free guidance scales increase prompt adherence at the cost of stochastic diversity [7]. Its fully documented training lineage makes it uniquely suited for mechanistic bias auditing, unlike later models such as SDXL [14] that rely on proprietary corpora. Zeng et al. [22] operationalise spurious feature analysis through a transformation-based classification framework in which each transformation, spanning pixel shuffle, patch shuffle, mean RGB, frequency filtering, Canny edge, SAM contour, monocular depth, semantic segmentation, object detection, and image captioning, preserves a targeted visual property while suppressing others, revealing that dataset identity is redundantly encoded across colour, texture, geometry, and semantic content simultaneously and that generative models can preserve or amplify training-data biases in their outputs. Meister et al. [12] provide a complementary occlusion-based probing protocol, establishing that gender artefacts are distributed across the entire image structure rather than localised to the subject, undermining fairness-through-blindness assumptions. This thesis extends Zeng et al.’s framework to a three-class Re-LAION-5B / SDv1.5 / COCO setting and applies Meister et al.’s occlusion protocol to both retrieved and generated corpora, extending it to skin tone as a second demographic dimension beyond Meister et al.’s gender-only scope.

Occupation prompts drawn from the ISCO-08 taxonomy [8] provide an interpretable probe for demographic priors [1, 10, 13]. Demographic annotation uses automated classifiers applied to segmented facial skin regions to prevent background shortcut learning [11], with skin tone operationalised via the ten-point Monk Skin Tone scale [20] and evaluated against CCv2 [15] and FACET [5]; automated annotation introduces structured measurement error [3]. Inference-time debiasing is applied via ITI-GEN [23], which learns inclusive text-space tokens grounded in demographic reference images, composed with ControlNet OpenPose [24] for structural conditioning and Chamfer-based colour alignment [21] to steer colour-level spurious priors.

3 Methodology

3.1 Image Retrieval and Dataset Construction

Re-LAION-2B-en-research-safe was processed across four randomly selected shards, yielding 65,552,809 URL-caption pairs of which 43,491,938 images were successfully retrieved (link decay rate 33.7%). CLIP ViT-L/14 embeddings were computed, L2-normalised, and indexed per shard using FAISS IndexFlatIP for exact cosine similarity retrieval. A set of 200 ISCO-08-aligned occupations was retrieved using the template "A photo of a/an {profession}" with gender-neutral phrasing, filtered via a negative prompt dominance criterion, and passed through a multi-stage structural validation pipeline requiring exactly one detected face, verified face-person spatial association, and successful MediaPipe FaceMesh landmark extraction, with images retained even where person detection was imperfect provided the face and FaceMesh criteria were satisfied. YOLOv12l-Face was selected as face detector (AP@0.5 0.905, highest small-face recall of four candidates). The top 1,000 structurally valid images per occupation were retained, yielding 200,000 images at a mean acceptance rate of 33.51%. A corresponding 200,000-image

generation corpus was produced using SDv1.5 with LCM-LoRA acceleration, selected for throughput (1.5 s/image versus 5.6 s for standard SDv1.5) from three LAION-traceable configurations. A full-body profession-conditioned prompt with neutral professional framing was used throughout; classifier-free guidance was fixed at $s=1.0$ to establish a conservative lower bound on bias free from inference-time amplification. Generated images passed through an analogous structural validation pipeline, achieving a mean acceptance rate of 66.38%.

3.2 Demographic Annotation

All images underwent FaceMesh face segmentation producing background-removed crops to prevent background shortcut learning. Gender was annotated using the Realistic Gender Classifier, selected from five candidates for its highest CCv2 face-only accuracy (88.6%) and smallest male-female gap among candidates with comparable overall accuracy (6.2 pp; the only candidate with a smaller gap sacrificed 11.5 pp overall accuracy). Skin tone was estimated using a VGG-16 MST10 regression model selected via factorial ablation across output formulation, colour space, label granularity, and background treatment, with predictions collapsed into Light (MST 1–3), Mid (MST 4–7), and Dark (MST 8–10) bins. All labels are treated as perceived-attribute proxies. Distributions were contextualised against BLS and ILOSTAT reference statistics; a race-to-skin-tone mapping via FairFace was abandoned due to mid-bin collapse.

3.3 Image Transformation Pipeline

The pipeline serves two objectives: dataset classification (ConvNeXt-Tiny distinguishing Re-LAION-5B, SDv1.5, and COCO per transformation) and demographic artefact probing (predicting gender and skin tone from isolated visual channels). Adding COCO as a third class enables per-class accuracy attribution, localising which dataset carries each spurious signal. Demographic probes use deliberately simple models: logistic regression for low-dimensional inputs and ResNet-50 for spatial inputs, following Meister et al.

Transformations span three levels. *Structural*: monocular depth (Depth-Anything-V2 ViT-S), Canny edge detection, low-pass and high-pass frequency filtering (hard circular FFT mask, $r=40$), and YOLOv8l-Pose keypoints (34-dimensional normalised vector). *Spatial-statistical*: pixel shuffle, patch shuffle at four scales (2×2 – 16×16), and mean RGB. *Semantic*: Mask2Former segmentation (Swin-Large, ADE20K), YOLOv8x object detection (OpenImages V7, demographic categories remapped), BLIP Large captions (gender-neutral remapped), and SDv1.5 VAE reconstruction. Occlusion variants follow Meister et al.’s five-condition protocol: Full NoBg, MaskSegm, MaskSegm NoBg, MaskRect, and MaskRect NoBg, with person masks from Mask2Former COCO (96.5% detection rate, highest of five candidates).

3.4 Inference-Time Debiasing Pipeline

A 10,000-image debiased corpus was generated for the doctor profession at $s=7.0$, fully activating the model’s demographic prior as a demanding test of the debiasing interventions. ITI-GEN [23] inclusive tokens were trained on CCv2 across gender and skin tone (MST 1–10), producing 20 demographic combinations of 500

images each, applied via the `prompt_prepend` mechanism. ControlNet OpenPose [24] (scale 0.6, fixed canonical skeleton) enforced structural consistency and reduced pose-correlated annotation confounds. Chamfer-based colour alignment [21] steered the predicted clean sample $\hat{x}_0$ at timesteps $t > 600$ toward a uniform mean-RGB reference derived from the standard SDv1.5 doctor corpus. LCM-LoRA was excluded due to incoherent outputs under per-step Chamfer modification; SW-Guidance was excluded on throughput grounds.

4 Evaluation

4.1 Demographic Distribution Analysis

Both datasets exhibit a pronounced male skew across 200 professions, but the imbalance is substantially more severe in SDv1.5. Re-LAION-5B contains 41.5% female-labelled images (58.5% male), a 17 percentage-point male–female gap; SDv1.5 widens this to 42 percentage points with only 29.02% female images. Against BLS 2024 and ILOSTAT baselines, Re-LAION-5B falls below real-world female proportions relative to US norms while remaining broadly consistent with global workforce proportions in several ISCO groups. SDv1.5 amplifies this skew across nearly every occupational group regardless of baseline, confirming a two-stage bias pattern in which the generative model exacerbates rather than merely inherits training corpus imbalances. At the ISCO-08 group level, this amplification occurs across every major group, with the strongest drop observed in Skilled Agricultural occupations, where the female proportion falls from 43.3% in Re-LAION-5B to 7.3% in SDv1.5. Gender amplification is confirmed in 178 of 200 professions, with extreme cases including Logger (97.5% male), Nurse (97.5% female), and Receptionist (97.3% female).

The skin tone analysis reveals a near-complete collapse of dark-toned representation in generative outputs: the Dark-bin proportion falls from 0.81% in Re-LAION-5B to 0.14% in SDv1.5 (83% reduction), while the Light-bin proportion nearly doubles from 12.45% to 22.87%. Skin tone amplification toward lighter tones is confirmed in 175 of 200 professions. A real-world comparison was not conducted as the FairFace-derived race-to-skin-tone mapping collapsed all racial groups to the Mid bin.

4.2 Dataset Classification

A ConvNeXt-Tiny classifier distinguishes Re-LAION-5B, SDv1.5, and COCO with high accuracy across every transformation, from unmodified images at 99.42% down to mean RGB at 64.19%. A consistent discriminating hierarchy emerges: frequency and fine spatial statistics (high-pass 98.64%, patch shuffle 8×8 at 97.91%); structural geometry (depth 96.78%, Canny edge 93.62%); dense semantics (segmentation 91.36%, captions 90.99%, object detection 88.62%); and global colour statistics (pixel shuffle 80.67%, mean RGB 64.19%). No single transformation eliminates separability, confirming that dataset identity is redundantly encoded across colour, texture, geometry, semantics, and language simultaneously. VAE reconstruction reduces accuracy by only 0.27 percentage points (99.15% vs. 99.42%), suggesting that low-level acquisition artefacts are unlikely to be the dominant source of separability. Accuracies substantially exceed the 82.0% reference reported by Zeng et al. for purely web-scraped datasets, consistent with SDv1.5’s generative pipeline imposing far

greater internal homogeneity than a web-scraped corpus. Table 1 summarises the corresponding test accuracy and AUC values.

Transformation	Test Acc. (%)	Test AUC
Base (unmodified)	99.42	0.9996
VAE reconstruction	99.15	0.9992
High-pass filter	98.64	0.9988
Patch shuffle 8×8	97.91	0.9983
Patch shuffle 16×16	97.69	0.9983
Low-pass filter	96.99	0.9969
Depth	96.78	0.9965
Patch shuffle 4×4	94.38	0.9949
Edge	93.62	0.9905
Semantic segmentation	91.36	0.9837
Captions [†]	90.99	0.9820
Patch shuffle 2×2	89.91	0.9766
Object detection	88.62	0.9728
Pixel shuffle	80.67	0.9430
Mean RGB	64.19	0.8207

[†] Logistic regression on BLIP Large captions encoded with SentenceT5-Base; all other rows use ConvNeXt-Tiny on rendered images.

Table 1: Dataset-source classification results for the three-class Re-LAION-5B / SDv1.5 / COCO task.

4.3 Demographic Artefact Probing

The occlusion experiments reveal a fundamental asymmetry. In Re-LAION-5B, all person-hidden variants collapse to chance for both gender and skin tone regardless of whether background is retained, establishing that background context carries no recoverable demographic signal. In SDv1.5, background context is a substantial independent carrier: with the person masked and background retained, gender AUC reaches 0.747 and skin tone AUC 0.728; silhouette alone (MaskSegm NoBg) reaches gender AUC 0.834. This is consistent with and substantially extends the contextual artefact identified by Meister et al.

Beyond occlusion, both datasets exhibit modest signal in non-appearance representations. Mean RGB achieves gender AUC 0.600/0.579 and skin tone AUC 0.563/0.543 (Re-LAION-5B / SDv1.5), consistent with the 0.580 reported by Meister et al. on COCO. Pose keypoints achieve gender AUC 0.697/0.680, confirming body configuration differences contribute independently of photometric properties. Object detection reveals gender-stereotyped scene compositions at AUC 0.692 on Re-LAION-5B (0.615 restricted), with both SDv1.5 object detection variants converging at AUC 0.608. Caption probing reveals a cross-dataset reversal absent from all visual probes: Re-LAION-5B achieves gender AUC 0.901 and skin tone AUC 0.749 versus SDv1.5 at 0.771 and 0.673, indicating that web-scraped images carry richer linguistically recoverable associations while SDv1.5 encodes stronger signal in scene-level visual statistics. Together, Re-LAION-5B shows no independent demographic signal in retained

background context, while person-level signal in isolation remains ambiguous under the applied masking protocol; SDv1.5, by contrast, distributes demographic signal pervasively across scene context, body geometry, object composition, and linguistic register. The full gender and skin tone separability results across transformations are summarised in Table 2.

Transformation	Re-L Gender	SD Gender	Re-L Skin	SD Skin
Base (unmodified)	0.986	0.946	0.968	0.900
Full, no background	0.499 [†]	0.948	0.514 [‡]	0.899
Masked rectangle	0.501 [†]	0.747	0.514 [‡]	0.728
Masked rectangle, no bg.	0.499 [†]	0.587 [§]	0.515 [‡]	0.660
Masked segmentation	0.501 [†]	0.852	0.514 [‡]	0.774
Masked segmentation, no bg.	0.501 [†]	0.834	0.515 [‡]	0.704
Depth at pose keypoints	0.559	0.592	0.598	0.655
Mean RGB	0.600	0.579	0.563	0.543
Object detection (restricted)	0.615	0.608	0.591	0.626
Object detection	0.692	0.608	0.611	0.663
Pose keypoints	0.697	0.680	0.596	0.666
Captions	0.901	0.771	0.749	0.673

[†] Gender AUC at chance level; model collapsed to majority-class prediction. [‡] Near-chance skin tone macro-AUC; per-class AUCs are at chance except MST8, affected by small-sample artefacts. [§] Near-chance AUC affected by near-complete majority-class prediction.

Table 2: Gender and skin tone separability AUCs across image transformations for Re-LAION-5B and SDv1.5. Gender uses binary AUC, while skin tone uses macro-averaged AUC over MST 1–8; chance level is 0.500.

4.4 Debiasing Evaluation

The combined ITI-GEN, ControlNet, and Chamfer colour alignment pipeline produces a substantive departure from the standard SDv1.5 demographic and spurious feature profile, achieving the intended 50% / 50% gender split and uniform spread across all ten MST levels. Person-level probing confirms successful conditioning: gender AUC reaches ≈ 1.000 on unmodified images (expected, as labels derive from the generation configuration).

Most non-person probes collapse to at or near chance, indicating substantial suppression of demographic cues in low-level, structural, object-level, and caption-based representations: mean RGB 0.507, object detection 0.449/0.457, depth at keypoints 0.454, pose keypoints 0.475, and captions 0.530, down 24.1 pp from the SDv1.5 value of 0.771. Skin tone non-person probes show the same pattern: mean RGB 0.482, object detection 0.439/0.461, depth 0.443, pose 0.483, and captions 0.490, all at or near the macro-AUC chance level of 0.500. The corresponding SDv1.5 and debiased probing results are summarised in Table 3.

However, background-retained occlusion (MaskRect) remains highly predictive (gender AUC 0.995; skin tone AUC 0.632), demonstrating that scene-level correlations persist despite debiasing.

The mean RGB skin tone value of 0.482 provides the clearest mechanistic evidence that Chamfer alignment substantially reduces photometric spurious correlations. The persistence of MaskRect performance indicates that scene composition remains partially entangled with demographic conditioning even under pose and appearance control, representing the primary open challenge for future work.

Representation	SD Gender	Deb. Gender	SD Skin	Deb. Skin
Base (unmodified)	0.946	≈ 1.000	0.900	0.861
Full, no background	0.948	0.998	0.899	0.823
Masked segmentation	0.852	0.999	0.774	0.729
Masked segmentation, no bg.	0.834	0.985	0.704	0.663
Masked rectangle	0.747	0.995	0.728	0.632
Masked rectangle, no bg.	0.587 [†]	0.701	0.660	0.543
Mean RGB	0.579	0.507	0.543	0.482
Object detection	0.608	0.449	0.663	0.439
Object detection (restricted)	0.608	0.457	0.626	0.461
Depth at pose keypoints	0.592	0.454	0.655	0.443
Pose keypoints	0.680	0.475	0.666	0.483
Captions	0.771	0.530	0.673	0.490

[†] Standard SDv1.5 value reflects near-complete majority-class collapse. Gender uses binary AUC. Skin tone uses macro-averaged AUC; SDv1.5 uses MST 1–8, while the debiased corpus uses MST 1–10. Chance level is 0.500.

Table 3: Gender and skin tone probing AUCs for standard SDv1.5 and the debiased corpus under the combined ITI-GEN, ControlNet OpenPose, and Chamfer colour alignment pipeline.

Taken together, the evaluation shows that demographic bias in SDv1.5 is not simply a direct copy of the imbalance present in Re-LAION-5B. Instead, the generative process transforms dataset-level demographic priors into a broader spurious-feature profile spanning background context, object composition, body geometry, colour statistics, and language. This distinction is important for mitigation: balancing generated demographic counts is necessary but insufficient if demographic information remains recoverable from non-person channels. The debiasing results show that inference-time conditioning suppresses many low-level and semantic shortcuts, but scene-level entanglement remains unresolved.

5 Conclusions and Future Work

This thesis applied and extended established spurious feature detection and demographic probing methodologies to Re-LAION-5B and Stable Diffusion v1.5, producing a systematic empirical characterisation of how demographic bias is encoded, distributed, and amplified across a web-scale training corpus and the generative model trained on it.

Addressing Objective 1, a profession-conditioned corpus of 200,000 images was retrieved from Re-LAION-5B and a corresponding 200,000-image generation corpus was produced using SDv1.5 with LCM-LoRA acceleration, both filtered through analogous structural validation pipelines. Demographic annotation was performed using the Realistic Gender Classifier and a VGG-16 MST10 Regression model evaluated against CCv2, MSTE, and FACET benchmarks, extending skin tone analysis to the full ten-point Monk Skin Tone scale beyond the coarser scales used in prior work. Addressing Objective 2, the transformation-based framework of Zeng et al. was extended to a three-class Re-LAION-5B / SDv1.5 / COCO configuration across fifteen transformations, with every transformation exceeding the 33.3% chance level, confirming that dataset-specific spurious signatures persist under extreme information compression and are redundantly encoded across colour, texture, geometry, semantics, and language simultaneously. Addressing Objective 3, demographic probing following the occlusion framework of Meister et

al. revealed a fundamental asymmetry between datasets: Re-LAION-5B exhibits demographic distributions that diverge from real-world labour force composition, with demographic signal absent from background context but present modestly across colour, geometry, and semantic representations. In contrast, SDv1.5 exhibits a fundamentally different structure: demographic information is distributed across scene context and background, recoverable from background context alone (gender AUC 0.747; skin tone AUC 0.728), from person silhouette geometry (gender AUC 0.834), and from natural language descriptions (gender AUC 0.771). Gender amplification was confirmed in 178 of 200 professions and skin tone amplification toward lighter tones in 175 of 200 professions. Addressing Objective 4, the combined ITI-GEN, ControlNet OpenPose, and Chamfer colour alignment pipeline successfully overrides SDv1.5's strongly skewed learned prior at the distributional level, achieving a uniform demographic distribution by construction with most non-person probes collapsing to at or near chance for both gender and skin tone. Residual occlusion signal persists, strongest when background is retained (gender MaskRect AUC 0.995; skin tone MaskRect AUC 0.632) but also detectable under MaskRectNoBg (gender AUC 0.701; skin tone AUC 0.543), indicating that scene composition and coarse person-region structure remain partially entangled with the demographic conditioning signal even under pose and appearance control. Addressing Objective 5, the combined evidence characterises a two-stage bias propagation-and-transformation process in which Re-LAION-5B exhibits demographic distributions that diverge from real-world labour force composition with no independent background-level demographic signal, and SDv1.5 then amplifies these imbalances substantially while transforming their distribution, projecting demographic priors beyond the person onto background, scene composition, object layout, and textual description. Inference-time debiasing can measurably reduce spurious correlations at the representation level without model retraining, but background-level entanglement represents the primary remaining challenge.

The most immediate future direction is targeted background conditioning to decorrelate scene composition from demographic attributes, whether through adversarial debiasing, extension of ITI-GEN to background properties, or weight editing via Unified Concept Editing [4], with a systematic comparison of conditioning-based versus weight editing approaches on the same probing protocol. Further extensions include evaluating the debiasing pipeline across a stratified sample of ISCO-08 major groups to assess generalisation beyond the doctor profession; validating debiased skin tone fidelity using automated MST estimation rather than generation-configuration labels; evaluating larger probe architectures or foundation model representations to determine whether the full-person/no-background ambiguity reflects masking limits or probe capacity; and extending the analysis to a larger shard sample of Re-LAION-5B to assess the stability of identified spurious feature profiles across the full corpus.

References

- [1] Federico Bianchi, Pratyusha Kalluri, Esin Durmus, Faisal Ladhak, Myra Cheng, Debora Nozza, Tatsunori Hashimoto, Dan Jurafsky, James Zou, and Aylin Caliskan. 2023. Easily Accessible Text-to-Image Generation Amplifies Demographic Stereotypes at Large Scale. In *Proceedings of the 2023 ACM Conference on Fairness, Accountability, and Transparency* (Chicago, IL, USA) (FAccT '23). Association for Computing Machinery, New York, NY, USA, 1493–1504. doi:10.1145/3593013.3594095
- [2] Abeba Birhane, Vinay Uday Prabhu, and Emmanuel Kahembwe. 2021. Multimodal datasets: misogyny, pornography, and malignant stereotypes. arXiv:2110.01963 [cs.CY] <https://arxiv.org/abs/2110.01963>
- [3] Joy Buolamwini and Timnit Gebru. 2018. Gender Shades: Intersectional Accuracy Disparities in Commercial Gender Classification. In *Proceedings of the 1st Conference on Fairness, Accountability and Transparency (Proceedings of Machine Learning Research, Vol. 81)*, Sorelle A. Friedler and Christo Wilson (Eds.). PMLR, 77–91. <https://proceedings.mlr.press/v81/buolamwini18a.html>
- [4] Rohit Gandikota, Hadas Orgad, Yonatan Belinkov, Joanna Materzyńska, and David Bau. 2024. Unified Concept Editing in Diffusion Models. arXiv:2308.14761 [cs.CV] <https://arxiv.org/abs/2308.14761>
- [5] Laura Gustafson, Chloe Rolland, Nikhila Ravi, Quentin Duval, Aaron Adcock, Cheng-Yang Fu, Melissa Hall, and Candace Ross. 2023. FACET: Fairness in Computer Vision Evaluation Benchmark. arXiv:2309.00035 [cs.CV] <https://arxiv.org/abs/2309.00035>
- [6] Jonathan Ho, Ajay Jain, and Pieter Abbeel. 2020. Denoising Diffusion Probabilistic Models. arXiv:2006.11239 [cs.LG] <https://arxiv.org/abs/2006.11239>
- [7] Jonathan Ho and Tim Salimans. 2022. Classifier-Free Diffusion Guidance. arXiv:2207.12598 [cs.LG] <https://arxiv.org/abs/2207.12598>
- [8] International Labour Organization. 2012. International Standard Classification of Occupations (ISCO-08). <https://ilostat.ilo.org/methods/concepts-and-definitions/classification-occupation/> Accessed: 2025.
- [9] LAION. 2024. Releasing Re-LAION-5B: transparent iteration on LAION-5B with additional safety fixes. <https://laion.ai/blog/relation-5b/>.
- [10] Alexandra Sasha Luccioni, Christopher Akiki, Margaret Mitchell, and Yacine Jernite. 2023. Stable Bias: Analyzing Societal Representations in Diffusion Models. arXiv:2303.11408 [cs.CY] <https://arxiv.org/abs/2303.11408>
- [11] Vitor Matias and João Neto. 2024. Enhancing Fairness in Machine Learning: Skin Tone Classification Using the Monk Skin Tone Scale. 76–81. doi:10.5753/sibgrapi.est.2024.31648
- [12] Nicole Meister, Dora Zhao, Angelina Wang, Vikram V. Ramaswamy, Ruth Fong, and Olga Russakovsky. 2023. Gender Artifacts in Visual Datasets. arXiv:2206.09191 [cs.CV] <https://arxiv.org/abs/2206.09191>
- [13] Ranjita Naik and Besmira Nushi. 2023. Social Biases through the Text-to-Image Generation Lens. arXiv:2304.06034 [cs.CY] <https://arxiv.org/abs/2304.06034>
- [14] Dustin Podell, Zion English, Kyle Lacey, Andreas Blattmann, Tim Dockhorn, Jonas Müller, Joe Penna, and Robin Rombach. 2023. SDXL: Improving Latent Diffusion Models for High-Resolution Image Synthesis. arXiv:2307.01952 [cs.CV] <https://arxiv.org/abs/2307.01952>
- [15] Bilal Porgali, Vitor Albiero, Jordan Ryda, Cristian Canton Ferrer, and Caner Hazirbas. 2023. The Casual Conversations v2 Dataset. arXiv:2303.04838 [cs.CV] <https://arxiv.org/abs/2303.04838>
- [16] Alec Radford, Jong Wook Kim, Chris Hallacy, Aditya Ramesh, Gabriel Goh, Sandhini Agarwal, Girish Sastry, Amanda Askell, Pamela Mishkin, Jack Clark, Gretchen Krueger, and Ilya Sutskever. 2021. Learning Transferable Visual Models From Natural Language Supervision. arXiv:2103.00020 [cs.CV] <https://arxiv.org/abs/2103.00020>
- [17] Robin Rombach, Andreas Blattmann, Dominik Lorenz, Patrick Esser, and Björn Ommer. 2022. High-Resolution Image Synthesis With Latent Diffusion Models. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*. 10684–10695.
- [18] Christoph Schuhmann. 2022. LAION-Aesthetics. <https://laion.ai/blog/laion-aesthetics/>.
- [19] Christoph Schuhmann, Romain Beaumont, Richard Vencu, Cade Gordon, Ross Wightman, Mehdi Cherti, Theo Coombes, Aarush Katta, Clayton Mullis, Mitchell Wortsman, Patrick Schramowski, Srivatsa Kundurthy, Katherine Crowson, Ludwig Schmidt, Robert Kaczmarczyk, and Jenia Jitsev. 2022. LAION-5B: An open large-scale dataset for training next generation image-text models. arXiv:2210.08402 [cs.CV] <https://arxiv.org/abs/2210.08402>
- [20] Candice Schumann, Gbolahan O. Olanubi, Auriel Wright, Ellis Monk Jr., Courtney Heldreth, and Susanna Ricco. 2024. Consensus and Subjectivity of Skin Tone Annotation for ML Fairness. arXiv:2305.09073 [cs.CV] <https://arxiv.org/abs/2305.09073>
- [21] Ka Chun Shum, Binh-Son Hua, Duc Thanh Nguyen, and Sai-Kit Yeung. 2025. Color Alignment in Diffusion. arXiv:2503.06746 [cs.CV] <https://arxiv.org/abs/2503.06746>
- [22] Boya Zeng, Yida Yin, and Zhuang Liu. 2024. Understanding Bias in Large-Scale Visual Datasets. arXiv:2412.01876 [cs.CV] <https://arxiv.org/abs/2412.01876>
- [23] Cheng Zhang, Xuanbai Chen, Siqi Chai, Chen Henry Wu, Dmitry Lagun, Thabo Beeler, and Fernando De la Torre. 2023. ITI-GEN: Inclusive Text-to-Image Generation. arXiv:2309.05569 [cs.CV] <https://arxiv.org/abs/2309.05569>
- [24] Lvmin Zhang, Anyi Rao, and Maneesh Agrawala. 2023. Adding Conditional Control to Text-to-Image Diffusion Models. arXiv:2302.05543 [cs.CV] <https://arxiv.org/abs/2302.05543>